



Breach Series

Part 3: Breach Trail

Centre For Cybersecurity Institute

Project Breach Trail

Third part of the breach series under the Windows Forensics track is focused on conducting a **automating the forensic investigation** of a compromised Windows system suspected to be part of a breach. Following alerts generated in *BreachTrace* and Red Team activity simulated in *Breach Point*, students will use forensic tools and techniques to acquire and analyze memory dumps, registry hives, log files, and user artifacts to reconstruct the attacker's activity and trace their digital footsteps.

Project Structure:

Scope 1

1. Automate Memory Analysis with Bash Script
 - 1.1. Check if the file can be analysed in Volatility; if yes, let the user choose to run Volatility2 or Volatility3.
 - 1.2. For both Volatility2 and 3:
 - 1.2.1. Find the memory profile and save it into a variable. (only for Volatility 2)
 - 1.2.2. Extract the Process Information into a file and Display the running processes.
 - 1.2.3. Extract the Network Connections into a file and Display network connections.
 - 1.2.4. Extract the Commands Executed into a file and Display the commands.
 - 1.2.5. Extract the Dll List into a file
 - 1.2.6. Extract hashes into a file and Display the hashes.
 - 1.2.7. Extract the Registry Information into a file
 - 1.2.8. Extract the SIDs into a file
2. Create a bash script that Automates HDD and Memory File Carving
 - 2.1. Check the current user; exit if not 'root'.
 - 2.2. Allow the user to specify the filename; check if the file exists.
 - 2.3. Create a function to install the forensics tools if missing.
 - 2.4. Use different carvers to automatically extract data.
 - 2.5. Data should be saved into a directory.
 - 2.6. Attempt to extract network traffic; if found, display to the user the location and size.
 - 2.7. Check for human-readable (exe files, passwords, usernames, IP Address, Emails, Darkweb activity etc.).
3. Documentation and Reporting:
 - 3.1. Display general statistics (time of analysis, number of found files, etc.).
 - 3.2. Save all the results into a report (name, files extracted, etc.)
 - 3.3. Zip the extracted files and the report file.

General Tools available: Bulk Extractor Binwalk Foremost Strings, Volatility.

Project Deliverables:

- Record the execution of the .sh to show how the script work and alerts generated into an .mp4 format.
- Use comments in your code to explain what you did.
- If you are using code from the internet, add credit and links
- Put the script (.sh), .mp4 and pdf report into a Zip File.
- Upload the .zip and recording to drive
- File Name: UNIT.STUDENT NUMBER (e.g. CCK1_240101.s5.zip, CCK1_240101.s5.sh, CCK1_240101.s5.pdf etc.)
- Failure to do so will be taken as no submission and a 0 grade for the project
- Upload the project to your individual google drive folder
- Last Submission: 23:59 of the project submission deadline. Folder will be denied access thereafter.